

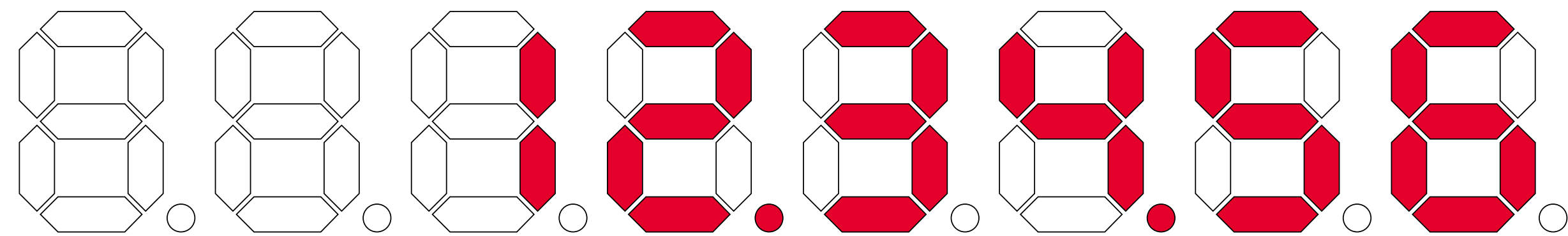


Abstract

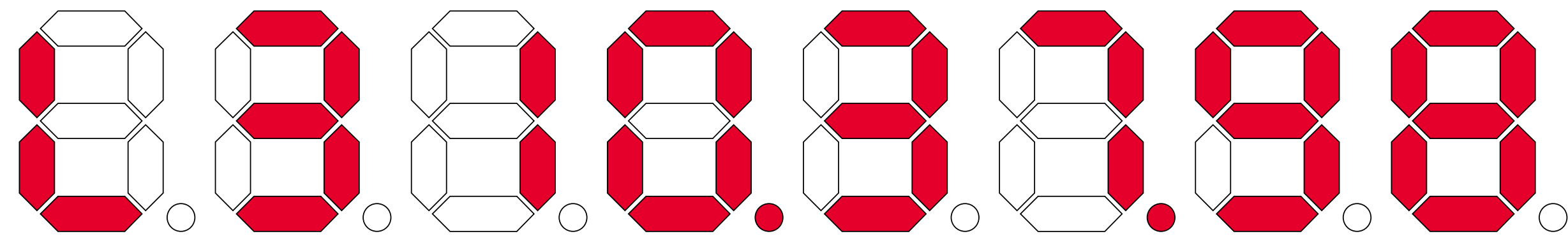
This group project was focused on implementing a digital stopwatch using the **Nexys A7 Artix-7 FPGA board**. The design and implementation was carried out using VHDL and AMD Vivado. The resulting design incorporates a counter based on the 100MHz clock signal of the development board with output on a seven-segment display. The stopwatch is controlled by 5 buttons arranged in a D-Pad style with the ability to save up to 9 laps into memory for later viewing.

Features

Mode 1: Displaying the time

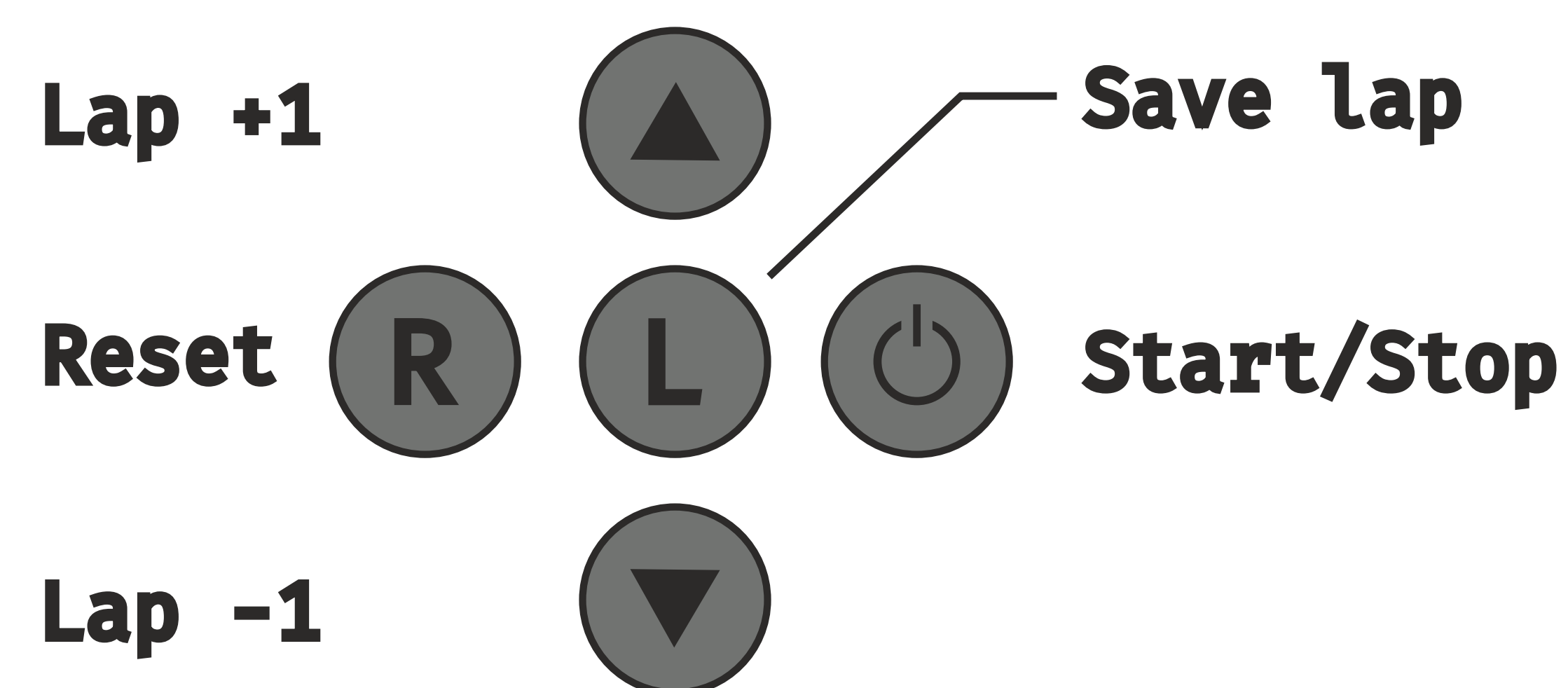


Mode 2: Displaying laps



The stopwatch is controlled using 5 buttons in a D-Pad configuration. It can display the current time being measured by the stopwatch component. The stopwatch counts with a precision to hundredths of a second and the format on the display is **mm:ss.ss**.

Using the Up and Down buttons the user can scroll between the 9 laps saved in the lap register. Laps are saved to the register using the Center button. Resetting the device is achieved via the Left button. Starting and stopping the stopwatch is done via the Right button.

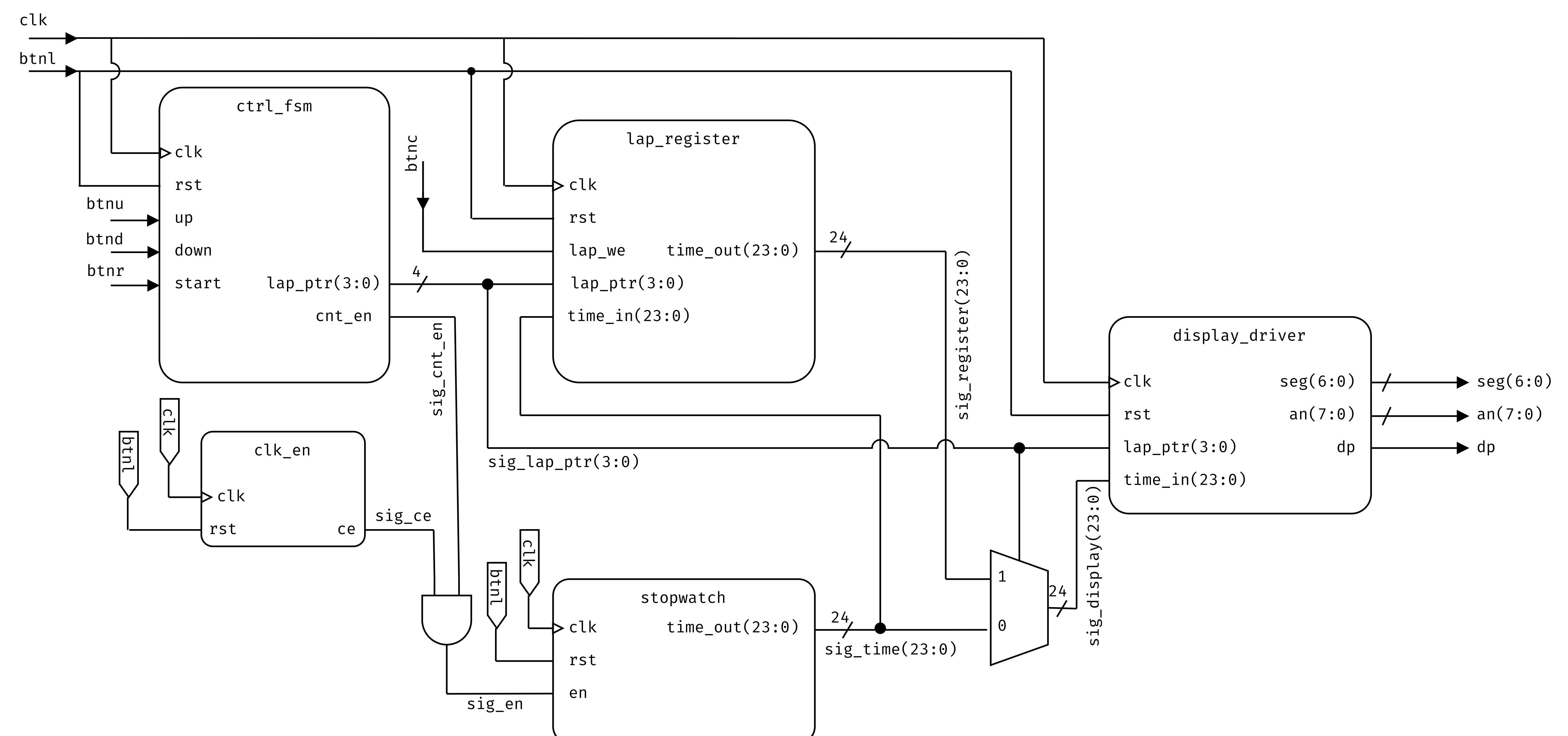


System design

The design follows a synchronous principle where all components trigger on the rising edge of the clock signal.

1. **ctrl_fsm**, this component handles starting and stopping the stopwatch. It also handles the navigation between saved laps using a pointer which tells the lap register which lap to show on its output.
2. **stopwatch**, made up of 6 cascading counter modules, all of which count a single digit in the resulting time. The 6 digits are in BCD format thus creating a 24-bit logic vector.
3. **lap_register**, is a register with space to hold 9 laps. The lap gets saved from the output of the stopwatch when the Center button is pressed. A lap pointer input tells the register which of the saved laps to output.
4. **display_driver**, is used to refresh the seven-segment displays and show the time data from its input. If the lap pointer is a non-zero number then it will simultaneously indicate which lap is being shown.

Top-level schema



Conclusion

We successfully implemented the design using VHDL and AMD Vivado. The synthesis and implementation steps executed with no errors and ended with utilising 331 LUTs, 491 Flip-Flops and 22 IO Blocks. No bugs were found when testing the implementation on real hardware.

References

- [1] VUT FEKT. Šablona pro psoter. https://latex.fekt.vut.cz/download/sabona_poster/, 2026.
- [2] Tomáš Frýza. vhd1-examples. <https://github.com/tomas-fryza/vhd1-examples>, 2026. Git repository. Accessed: 04-05-2026.
- [3] Wikimedia Commons. 7 segment display, svg image. https://commons.wikimedia.org/wiki/File:7_Segment_Display.svg, 2006.